Tanuj Dargan

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Education & Awards

University of Victoria – International Scholarship Victoria, BC
B.S. Computer Science – Accelerated 3-Year Completion *Apr. 2027*
Relevant Coursework: Quantum Computing, Computer Architecture, Numerical Analysis, Communications & Networks

Work Experience

Pear Care Jul. 2025 – Present
Lead AI Developer *Remote*

- **Accelerated diagnostic workflows** by building autonomous medical-AI agents with LangGraph, MoE routing, and task-introspective reasoning, served via FastAPI on GCP across 1,000+ clinical specialties.
- **Cut model memory footprint by 40%** while preserving 95%+ F1 on internal clinical triage benchmarks by self-hosting with PyTorch, Ollama, and vLLM, then transitioning to quantized LoRA/PEFT fine-tuned deployments.
- **Architected for horizontal scale** across a multi-tenant GCP/AWS infrastructure with Redis caching, DynamoDB persistence, end-to-end encryption, and mobile-first access, validated through load and stress testing.

Major League Hacking (MLH) SWE Fellowship - Apache Airflow May 2025 – Aug. 2025
Software Engineering Fellow *Remote*

- **Streamlined developer debugging for Apache Airflow's 50K+ star codebase** by architecting Docker-based IDE integration (VSCode, PyCharm) with step-through debugging and dynamic port exposure, cutting setup-to-debug cycles from hours to minutes; shipped in a team of 3 alongside Royal Bank of Canada engineers.
- **Diagnosed and patched a long-standing time-window logic bug** causing silently missed DAG runs across multiple Airflow releases; fix merged into the core scheduler.

DEIA (Database & Data Mining) May 2025 – Present
Undergraduate Researcher - Prof. Sean Chester *Victoria, BC*

- **Sustained 1–4M QPS at billion-vector scale** by building a custom retrieval system on a modified cuVS fork (CAGRA, IVF-PQ) with 90–95% recall and sub-250 ms latency across 8+ NVIDIA GPUs and 2TB of indexed data.
- **Achieved 4.3× throughput scaling** by building automated stress-testing pipelines on Kubernetes GPU clusters, profiling memory bandwidth and NVLink interconnect saturation to expose bottlenecks.

SOLIDS (Software for Large-Scale, Distributed, Intelligent, and Secure Systems) May 2025 – Present
Undergraduate AI Researcher - Dr. Navneet Popli *Victoria, BC*

- **Achieved $R^2 = 93.6\%$ satellite-to-deep-water temperature prediction** (+17.7% over baseline) across 9 experiments, disproving the original 167 m-depth hypothesis and pivoting the research direction for the lab.
- **Delivered and deployed 3 ocean-health frameworks** (SHQI, CVI, OHI at $R^2 = 87.9\%$) to the National Research Council Canada, now in active use for salmon habitat monitoring, coastal erosion early warning, and First Nations stewardship dashboards.

UVic Centre for Aerospace Research (CFAR) Apr. 2025 – Present
Software Developer & Undergraduate Researcher *Victoria, BC*

- **Led OBC firmware (C), TTC board design, and cross-functional hardware/software integration** for DoomSat and a high-altitude balloon mission, coordinating across electrical, mechanical, and software teams.
- **Cut MarmotSat testing from 180 min to 15 min** by building a CI/HIL pipeline with GitHub Actions and SWD flashing (80+ checksum rules/commit), and implementing C/Java YAMCS integration that eliminated 20+ hours of manual work per release.

Projects

Hive Agent Framework – Open Source Contributor | *7.7K+ Stars, Python, Bash* Jan. 2026 – Present

- **Resolved a cross-platform setup blocker** (Issue #476) by overhauling Python detection with multi-candidate probing, timeout handling, and `pipefail` hardening across Windows, macOS, and Linux; actively contributing additional tooling.

Optimate: Hack the North 2025 Winner | *Next.js, Cohere, AWS DynamoDB* Sep. 2025

- **Won 2 prizes out of 300+ teams** (**YC Unicorn Prize** & **Federato Best RiskOps**) by building an AI underwriting dashboard that automated risk assessment with Next.js, Cohere LLMs with RL, and DynamoDB; Federato CEO validated 40% faster review cycles and 15% fewer underwriting errors versus manual workflows.

AIMO Progress Prize 2 Competitor: Silver Medal | *Finishing score: 27* Nov. 2024 – Mar. 2025

- **Achieved 54% accuracy, finishing top 10% of 2,200 teams** by engineering multi-stage LLM pipelines with advanced prompt engineering, modular reasoning strategies, and hyper-parameter tuning.

Technical Skills

Languages: Python, C/C++, Java, JavaScript/TypeScript, Go, Bash, SQL
AI/ML: PyTorch, vLLM, Ollama, LangGraph, LoRA/PEFT, Hugging Face, OpenCV, Pandas, cuVS
Frameworks & Tools: FastAPI, Next.js, React, Node.js, Flutter, Django, Flask, Docker, Kubernetes, Git, GitHub Actions
Infrastructure: AWS (EC2, S3, Lambda, SageMaker, DynamoDB, Bedrock), GCP, Redis, Proxmox, Nginx, Cloudflare, Slurm